

ZONGULDAK, TurkeyWMO: **170220**Lat: **41.4495N**Lon: **31.7785E**Elev: **137**StdP: **99.69**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	-0.1	-6.8	2.1	3.5	-5.3	2.5	3.7	8.8	8.7	7.7	8.5	3.0	180	0.360

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	28.5	21.0	27.4	21.6	26.5	21.4	24.1	26.8	23.3	26.1	22.6	25.5	2.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	18.3	26.0	22.3	17.3	25.5	21.6	16.5	24.9	73.4	26.8	70.2	26.2	67.5	25.6	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.9	5.8	5.0	-2.5	32.6	1.7	2.4	-3.8	34.4	-4.8	35.8	-5.8	37.2	-7.1	38.9		
(5)			-3.5	25.0	1.7	1.2	-4.8	25.9	-5.8	26.6	-6.7	27.3	-8.0	28.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	6.5	6.8	8.5	11.9	16.0	20.0	22.5	22.8	19.7	15.7	11.9	8.4
	DBStd	6.81	4.27	4.29	4.62	4.45	3.56	2.48	1.77	1.75	2.52	3.39	3.90	4.35
	HDD10.0	458	126	108	89	26	1	0	0	0	0	2	23	84
	HDD18.3	1914	366	322	308	200	92	10	0	0	14	96	197	309
	CDD10.0	2013	18	19	41	84	187	300	388	396	291	177	79	34
	CDD18.3	428	0	0	1	8	19	60	130	137	56	13	3	1
	CDH23.3	1863	0	1	8	43	88	213	612	681	167	39	7	4
CDH26.7	204	0	0	1	10	15	33	59	58	22	6	0	0	
Wind	WSAvg	2.1	2.3	2.3	2.3	2.0	1.9	1.9	2.1	2.1	2.1	2.0	2.1	2.3
Precipitation	PrecAvg	1197	131	92	95	59	58	71	75	85	108	141	142	161
	PrecMax	1666	250	205	180	155	230	228	261	341	353	324	360	277
	PrecMin	822	4	15	18	5	11	17	4	0	1	15	9	41
	PrecStd	168	58	43	41	32	39	46	68	80	86	69	78	55
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	20.6	24.0	28.1	28.4	30.1	30.9	29.7	29.0	26.6	24.3	23.2
		MCWB	10.3	11.7	12.9	15.0	17.3	19.8	21.0	20.9	19.5	17.2	14.4	12.1
	2%	DB	16.6	17.5	20.1	23.9	25.8	27.0	28.0	27.9	26.6	24.0	21.1	19.2
		MCWB	9.5	10.0	11.8	14.3	17.0	19.9	22.6	23.4	20.0	16.7	13.4	11.5
	5%	DB	14.9	15.3	17.7	21.1	23.5	25.5	27.0	27.1	25.0	22.0	19.2	17.0
		MCWB	8.8	9.1	10.6	13.3	16.2	19.8	22.5	22.8	20.0	17.1	12.7	10.4
	10%	DB	13.1	13.4	15.7	18.5	21.5	24.2	26.1	26.4	23.9	20.3	17.5	15.0
		MCWB	8.0	8.3	9.6	12.0	15.6	19.6	21.9	22.3	19.5	16.7	12.2	9.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	12.5	14.2	16.4	19.1	23.2	24.8	25.8	22.8	19.9	16.1	13.6
		MCDB	17.0	18.0	21.8	25.3	24.3	26.0	27.2	27.4	25.7	22.3	19.6	19.6
	2%	WB	10.2	10.8	12.3	14.7	18.1	22.0	24.0	24.5	21.7	18.8	14.8	12.2
		MCDB	15.0	15.8	18.6	22.1	22.6	24.6	26.8	26.9	24.6	21.3	19.3	17.6
	5%	WB	9.2	9.7	11.1	13.6	17.3	21.1	23.3	23.6	20.9	17.9	13.8	11.1
		MCDB	13.5	14.4	16.7	19.6	21.5	24.0	26.1	26.3	23.7	20.6	17.6	15.7
	10%	WB	8.4	8.7	10.0	12.6	16.5	20.2	22.6	22.9	20.1	17.0	12.9	10.1
		MCDB	12.3	13.0	15.2	17.4	20.4	23.3	25.5	25.7	23.0	19.7	16.4	14.1
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.1	5.5	5.8	5.7	6.1	6.3	6.2	6.2	5.5	5.4	5.1
		MCDBR	7.3	8.3	9.1	9.8	8.6	7.7	7.0	6.9	7.5	7.4	7.7	8.0
		MCWBR	3.8	4.2	4.4	4.5	4.8	5.0	5.0	5.0	5.2	4.4	4.1	4.3
	5% WB	MCDBR	6.5	7.8	8.5	8.3	6.9	6.4	6.2	6.3	6.7	6.1	6.5	7.0
		MCWBR	3.7	4.4	4.4	4.4	4.4	4.7	4.8	4.9	5.0	4.1	4.1	4.1
Clear-Sky Solar Irradiance	taub	0.351	0.380	0.418	0.453	0.426	0.410	0.417	0.422	0.408	0.390	0.360	0.350	
	taud	2.357	2.269	2.178	2.112	2.270	2.361	2.356	2.338	2.360	2.374	2.403	2.383	
	Ebn at Noon	783	813	822	820	854	866	855	840	826	795	769	751	
	Edh at Noon	86	108	133	152	133	122	121	120	110	97	82	78	
All-Sky Solar Radiation	RadAvg	1.43	2.20	3.42	4.77	6.01	6.86	7.06	6.18	4.57	3.02	1.99	1.36	
	RadStd	0.19	0.31	0.29	0.49	0.52	0.52	0.29	0.41	0.35	0.27	0.25	0.19	